In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
df=pd.read_csv("powerplant_data.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	AT	V	AP	RH	PE
0	8.34	40.77	1010.84	90.01	480.48
1	23.64	58.49	1011.40	74.20	445.75
2	29.74	56.90	1007.15	41.91	438.76
3	19.07	49.69	1007.22	76.79	453.09
4	11.80	40.66	1017.13	97.20	464.43

In [4]:
```
#AT=TEMPARATURE
#V=VACCUM
#AP=PRESSURE
#RH=HUMIDITY
#PE=PRODUCED ENERGY
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
AT    0
V     0
AP    0
RH    0
PE    0
dtype: int64
```

In [6]:
```
# PE = Target (Output)
# Remaining columns = Inputs (Features)
X=df.drop("PE",axis=1)
y=df["PE"]
y
```

Out[6]:
```
0       480.48
1       445.75
2       438.76
3       453.09
4       464.43
         ...
9563    462.59
9564    432.90
9565    465.96
9566    450.93
9567    451.67
Name: PE, Length: 9568, dtype: float64
```

In [7]:
```
#SPLIT OUT DATA
from sklearn.model_selection import train_test_split

X_train,X_test,y_train,y_test=train_test_split(
```

```
    X,y,test_size=0.2,random_state=42
)
```

In [8]:
```
#Scaling
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
X_train_scaled=scaler.fit_transform(X_train)
X_test_scaled=scaler.transform(X_test)
```

DATA TO TENSOR

In [1]:
```
# | Part    | Meaning                |
# | ------- | ---------------------- |
# | `torch` | Main PyTorch library   |
# | `nn`    | Neural Network module  |
# | `as nn` | Short name (alias)     |

#  Why do we use torch.nn?
# Because it provides ready made neural network components:
#  Layers
#  Loss functions
# Activation functions
#  Models
#Without this module, building ANN would be very difficult.
```

In [2]:
```
# 🔷 .values
# Converts pandas column → NumPy array.

# 🔷 .view(-1,1)
# Changes shape
# BEFORE
# [10,20,30,40]
# shape = (4,)
# AFTER
# [[10],
#  [20],
#  [30],
#  [40]]
# shape = (4,1)
# .view() changes the SHAPE of a tensor without changing data.
# Data same
# Arrangement changes
```

In [11]:
```
import torch
import torch.nn as nn
X_train_tensor=torch.tensor(X_train_scaled,dtype=torch.float32)
y_train_tensor=torch.tensor(y_train.values,dtype=torch.float32).view(-1,1)

X_test_tensor=torch.tensor(X_test_scaled,dtype=torch.float32)
y_test_tensor=torch.tensor(y_test.values,dtype=torch.float32).view(-1,1)
```

TensorDataset and DataLoader

In [12]:
```
from torch.utils.data import TensorDataset,DataLoader
#Dataset
train_dataset=TensorDataset(X_train_tensor,y_train_tensor)
test_dataset=TensorDataset(X_test_tensor,y_test_tensor)
```

In [13]:
```
#DataLoader
train_loader=DataLoader(train_dataset,batch_size=32,shuffle=True)
test_loader=DataLoader(test_dataset,batch_size=32)
```

Deep Learning (Building ANN Model)

In [14]:
```
#Basics
#  What is __init__ ?
# Definition
# __init__ is a special function in Python that runs automatically when an objec
# It is called a constructor.

# class Student:
#     def __init__(self, name):
#         self.name = name
#         #Create object:
#         s1 = Student("Raj")
#         #Python internally does:
#         Student.__init__(s1,"Raj")

#What is self ?
#self = current object
# It represents the object being created.

# Think Like This

# If:
# s1 = Student("Raj")
# then:
# self = s1

# If:
# s2 = Student("Amit")
# then:
# self = s2
```

Define our ANN model

In [15]:
```
# nn.Module = Car Engine
# ANN = Car Body
# super() = Installing Engine

class ANN(nn.Module):#ANN inherits from nn.Module(so nn parent and ANN is child)
    def __init__(self):
        super(ANN,self).__init__()#It initializes parent class (nn.Module).

        #What is Sequential?:-It connects layers one after another automatically
        self.model=nn.Sequential(
            #1st Hidden layer
```

```
            nn.Linear(X_train.shape[1],6),
            nn.ReLU(),

            #2ND Hidden layer
            nn.Linear(6,6),
            nn.ReLU(),

            #Output layer
            nn.Linear(6,1),
        )
    #Forward passs(Data movement through network.)
    def forward(self,x):
        return self.model(x)
```

In [16]:
```
#Model Object Creation and loss function and optimizer
# ANN object created
#         ↓
# __init__ runs
#         ↓
# Layers initialized
#         ↓
# Model ready
import torch.optim as optim
model=ANN()

#Loss,Optimizer
criterion=nn.MSELoss()
optimizer=optim.Adam(model.parameters())
```

Training Our Model(ANN)

In [17]:
```
# For each epoch:
#     Set running_loss = 0

#     For each batch:
#         Forward
#         Compute loss
#         Backward
#         Update weights
#         Add loss to running_loss

#     After all batches:
#         Divide running_loss by total batches
#         Store average as epoch_train_loss
```

In [18]:
```
# running_loss = collects batch losses
# epoch_train_loss =gives average loss of whole dataset for that epoch
train_losses=[]
val_losses=[]

best_val_loss = float("inf")#Intially (infinity)

epochs=100
for epoch in range(epochs):
    model.train()
    running_loss=0.0#total training loss for 1 epoch
```

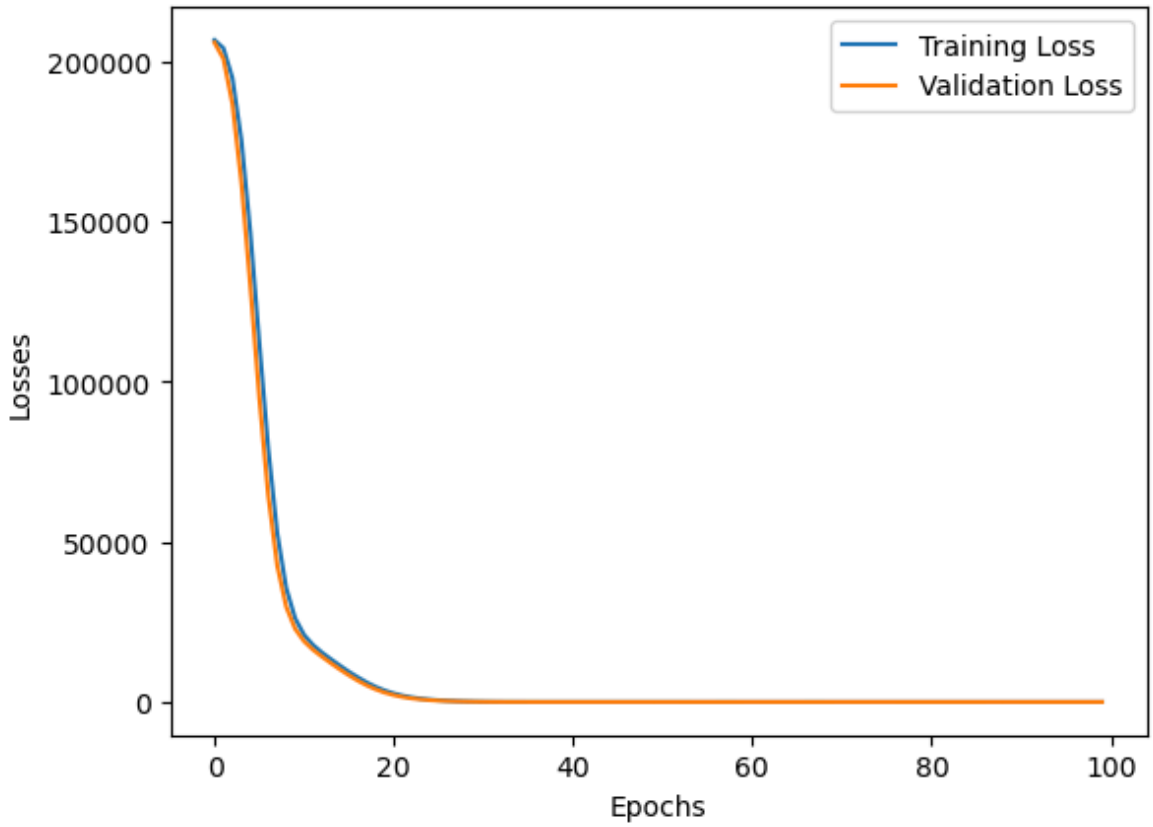
```
    for xb,yb in train_loader:
        #xb=features of 1 batch
        #yb=labels(output feature) of 1 batch
        optimizer.zero_grad()#clears (resets) all previously stored gradients be

        outputs=model(xb)#forward prop......predicted outputs for this batch
        loss=criterion(outputs,yb)#Compute loss(predicted,actual)
        loss.backward()#back prop....compute gradients
        optimizer.step()#paramters update

        running_loss+=loss.item()#loss is a tensor ->py float

    epoch_train_loss=running_loss/len(train_loader)
    train_losses.append(epoch_train_loss)

    #Validation(During this no backward propgation because we are validating(tes
    model.eval()
    running_val_loss=0.0

    with torch.no_grad():#No gradients computed
        for xb,yb in test_loader:
            outputs=model(xb)
            loss=criterion(outputs,yb)
            running_val_loss+=loss

    epoch_val_loss=running_val_loss/len(test_loader)
    val_losses.append(epoch_val_loss)

    if epoch_val_loss<best_val_loss:
        best_val_loss=epoch_val_loss
        torch.save(model.state_dict(),"best_model.pt")#.pt or .pth
```

In [19]:

```
#Visualize the loss
import matplotlib.pyplot as plt

loss_df = pd.DataFrame({
    "Training Loss":train_losses,
    "Validation Loss":val_losses
})

plt.plot(loss_df["Training Loss"], label = "Training Loss")
plt.plot(loss_df["Validation Loss"], label = "Validation Loss")

plt.xlabel("Epochs")
plt.ylabel("Losses")

plt.legend()
```

Out[19]:

```
<matplotlib.legend.Legend at 0x243d71fa230>
```



Loading the best model

In [20]:
```
# What is happening internally?

# torch.load("best_model.pt")
# → Loads the saved parameter dictionary (weights & biases).

# model.load_state_dict(...)
# → Puts those saved weights into your model.
#"Load the best learned brain into my model."
```

In [21]:
```
model.load_state_dict(torch.load("best_model.pt"))
```

```
C:\Users\Vedant\AppData\Local\Temp\ipykernel_8684\349716640.py:1: FutureWarning:
You are using `torch.load` with `weights_only=False` (the current default value),
which uses the default pickle module implicitly. It is possible to construct mali
cious pickle data which will execute arbitrary code during unpickling (See http
s://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more de
tails). In a future release, the default value for `weights_only` will be flipped
to `True`. This limits the functions that could be executed during unpickling. Ar
bitrary objects will no longer be allowed to be loaded via this mode unless they
are explicitly allowlisted by the user via `torch.serialization.add_safe_globals
`. We recommend you start setting `weights_only=True` for any use case where you
don't have full control of the loaded file. Please open an issue on GitHub for an
y issues related to this experimental feature.
  model.load_state_dict(torch.load("best_model.pt"))
```

Out[21]:
```
<All keys matched successfully>
```

Evaluation

In [23]:
```
model.eval()
with torch.no_grad():
    train_preds=model(X_train_tensor)
    test_preds=model(X_test_tensor)

    train_mse_loss=criterion(train_preds,y_train_tensor)
    test_mse_loss=criterion(test_preds,y_test_tensor)

print("Training MSE:", train_mse_loss.item())
print("Testing MSE:", test_mse_loss.item())
```

```
Training MSE: 20.67180061340332
Testing MSE: 19.034019470214844
```

In [27]:
```
from sklearn.metrics import r2_score
print("r^2 score =", r2_score(y_test, test_preds))
```

```
r^2 score = 0.9334810172481811
```

In [28]:
```
#Predicted vs actual value
predicted_df = pd.DataFrame(test_preds.numpy(), columns=["Predicted Values"])
actual_df = pd.DataFrame(y_test.values, columns=["Actual Values"])

pd.concat([predicted_df, actual_df], axis=1)
```

Out[28]:

	Predicted Values	Actual Values
0	435.509277	433.27
1	436.978333	438.16
2	461.587067	458.42
3	476.497986	480.82
4	435.551422	441.41
...	...	...
1909	451.524994	456.70
1910	431.767090	438.04
1911	467.888123	467.80
1912	431.166840	437.14
1913	456.657928	456.78

1914 rows × 2 columns